Scutoid Tissues in Four Dimensions

the boundary of every simple 4-polytope is a closed scutoid tissue whose moves are counted by its h-vector; the Swap Rule of close packing below s = 1/4, whose walls are rosette exchanges



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DICTO: ideas, names, constructions and direction.
Computation, verification and drafting with Claude (Anthropic).

Version 1. Theorems 1, 1′, 2 and 4 and Lemma 5 are proved here; their counts are also confirmed by computer sweeps (floating point), each predicted before counting. Theorem 3 is an exact computer-assisted proof for every layer spacing in [1/8, 1/4] in the six stackings that contain all sixteen four-layer contexts; for other stackings it assumes that a piece is fixed by the four layers round its gap; the region below $\lambda = -9s/8$ and $s < 1/8$ are open. Checked before release by four automated referee passes and one independent attack (reports and judgements kept with the study data).

A scutoid is a cell between two surfaces whose outline changes on the way through: two cells that touched below stop touching, and two others start. Gómez-Gálvez and co-workers found the shape in curved living tissue in 2018; they called the change an apico-basal transition and compared it to the T1 transition of foams and cell sheets (in a 2D cell pattern a T1 is the classical 2-2 flip of the dual triangulation: Weaire and Rivier; Weaire and Hutzler). We find such tissues where one would not look: on the boundary of a four-dimensional polytope. Slice a 4-polytope with a moving hyperplane; its 3-dimensional cells cross the slice as polygons tiling the slice's surface, and as the hyperplane moves the polygons trade neighbours. Each cell, followed through the sweep, is a flat-faced scutoid in the 2018 sense, and the boundary is a closed scutoid tissue. For simple 4-polytopes the number of neighbour exchanges is fixed by the h-vector, in every direction; for the polytope halfway between the 600-cell and the 120-cell they balance exactly between its two kinds of cell. Then one dimension up (Section 3): the holes of close packing, their power cells stacked over a weight λ, are 4-dimensional cells whose 3D slices exchange neighbours at the walls of the Swap Rule (#61), there in symmetric clumps of the kind biologists call rosettes.

1. The boundary of a simple 4-polytope is a scutoid tissue

A 4-polytope is *simple* if each corner lies in exactly four cells (equivalently, four edges). Take a generic height, a linear function with different values at all corners. Slicing at height t gives a 3-polytope whose faces are the cross-sections of the cells that cross the hyperplane; two cells are neighbours in the slice when their shared 2-face crosses it. At a simple corner the faces

through it correspond to the subsets of its four edges, and a face has the corner as its top iff all its edges there go down.

Theorem 1. In a generic sweep of a simple 4-polytope the tiling of the slice by the cells changes only when the hyperplane passes a corner, and then in exactly one of three ways, according to the number i of the corner's edges that go down: i = 1, one cell is born (a small triangle appears); i = 2, a **scutoid swap**: one pair of cells stops touching and one pair starts, nothing else; i = 3, one cell dies. The numbers of births, swaps and deaths are h_1, h_2 and h_3, the middle entries of the polytope's h-vector, the same for every generic direction.

Proof. Between corner heights the slice keeps its combinatorics. At a corner v let $e_1, \dots, e_4$ be its edges and F_j the cell not containing e_j. Just below v the slice cuts the downward edges, just above the upward ones. i = 1 (e_1 down): above v the slice cuts e_2, e_3, e_4, and the cell containing all three, F_1, appears as a small triangle (v is its lowest corner). i = 2 (e_1, e_2 down): below v the slice's local edge joins the crossings of e_1 and e_2 and lies in $F_3 \cap F_4$, the 2-face spanned by e_1 and e_2; v is its top, so above v the two cells no longer share a face crossing the hyperplane (two cells of a polytope share at most one 2-face); symmetrically $F_1 \cap F_2$, spanned by e_3, e_4, has v as its bottom and F_1, F_2 begin to touch. No other face through v has v as top or bottom, so nothing else changes. i = 3 is i = 1 upside down. The number of corners with i downward edges is h_i, independent of the generic height (the Morse-type count of the h-vector, with $h_i = h_{4-i}$; McMullen; Ziegler, Lectures on Polytopes, §8.2; Buchstaber–Panov, Toric Topology, ch. 1). ∎ In the dual triangulation of the slice's boundary (one vertex per cell) the three events are the bistellar flips 1-3, 2-2 and 3-1 (Pachner; Lickorish); the theorem is classical in that language. What it adds is the reading: the cells are scutoids, the 2-2 flips are their neighbour exchanges, and h_2 counts them.

Theorem 1′ (every dimension). In a generic sweep of a simple d-polytope, two facets touch in the slice exactly when their common ridge crosses the hyperplane. Passing a corner with i downward edges is the bistellar $i \to (d - i)$ move on the dual triangulation of the slice's boundary, and it changes the facets' contacts only for i = 1 (a facet is born), i = 2 (one contact is gained), $i = d - 2$ (one contact is lost) and $i = d - 1$ (a facet dies). For d = 4, $i = 2 = d - 2$ and the gain and the loss happen together (the swap of Theorem 1); for d = 5 they are separate moves at i = 2 and i = 3; for $d \geq 6$ the corners with $3 \leq i \leq d - 3$ change no facet contact (they move lower faces only). The number of corners of index i is h_i.

Proof. A ridge $F_a \cap F_b$ through v is spanned by the $d - 2$ edges other than e_a, e_b; it has v as its top iff they all go down (so $i = d - 2$ and {a, b} are the up edges), as its bottom iff they all go up (i = 2); otherwise it crosses on both sides. ∎ *Confirmation* (floating point, predictions first, three random directions each): the 5-cube, h = (1, 5, 10, 10, 5, 1): 5 births, 10 contacts gained, 10 lost, 5 deaths; the 5D permutohedron (the 720 orderings of 1, …, 6), h = the Eulerian numbers (1, 57, 302, 302, 57, 1): 57, 302, 302, 57; the 4D permutohedron, (1, 26, 66, 26, 1): 26 births, 66 swaps, 26 deaths. **No regularity is needed:** three random simple 4-polytopes (polars of hulls of 24, 36, 54 random points near the 3-sphere, no symmetry) gave exactly the predicted (20, 57, 20), (32, 103, 32), (50, 142, 50). *Remark.* At a corner of a non-simple polytope the moves clump; a generic small shift of the facet hyperplanes makes any polytope simple and splits each clump into births, swaps and deaths.

Theorem 2 (the polytopes between the Queens). On the path of uniform truncations from the 600-cell (the White Queen) to the 120-cell (the Red Queen), the 120-cell, the truncated 120-cell and the bitruncated 600-cell are simple, with (births, swaps, deaths) = (116, 366, 116), (716, 966, 716) and (716, 2166, 716) in every generic sweep.

Proof. Their vertex figures are tetrahedra; $h_1 = f_3 - 4$ and $h_2 = f_0 - 2 - 2h_1$ follow from the Dehn–Sommerville relations; then Theorem 1. *Confirmation:* computer sweeps along generic directions give the predicted counts (the bitruncated 600-cell along nine directions). ∎

Lemma 5 (the truncated 600-cell has no pure swap). At every corner of the truncated 600-cell, in every direction, at least three contacts change at once.

Proof. Six cells meet at each corner (one icosahedron and five truncated tetrahedra); the vertex figure is a pentagonal pyramid, whose graph is the wheel W_5 with 10 edges. A pure swap needs the down/up split of the vertex figure's vertices to leave exactly one edge inside each side; but the largest cut of W_5 has 7 edges (the hub with two non-adjacent rim vertices), so at least three edges lie inside the two sides, each a contact that is lost or gained. ∎ In five computer sweeps the moves were $1 \leftrightarrow 3$, $2 \leftrightarrow 0$ and $1 \leftrightarrow 0$ clumps. Biologists know the same contrast in living tissue: a T1 transition involves four cells at a point, while a *rosette*, five or more cells meeting at one point and resolving together, reshuffles several neighbours at once (Blankenship, Backovic, Sanny, Weitz and Zallen 2006). In that language the truncated 600-cell, with its groups of five truncated tetrahedra round each icosahedron's corner, is a rosette tissue, and the bitruncated 600-cell, one stage on, a T1 tissue.

polytope	corners	cells	cells per corner	births	scutoid swaps	deaths
120-cell (Red Queen)	600	120 dodecahedra	4	116	366	116
truncated 120-cell	2400	120 truncated dodecahedra + 600 tetrahedra	4	716	966	716
bitruncated 600-cell (the midpoint)	3600	120 truncated icosahedra + 600 truncated tetrahedra	4	716	2166	716
truncated 600-cell	1440	120 icosahedra + 600 truncated tetrahedra	6	rosette moves only: ≥ 3 contacts change at every corner (Lemma 5)		

In symmetric directions many corners share a height and the events come in batches; along cell-first the bitruncated 600-cell's pairs that let go and pairs that meet are equal in total (1782 each) and at every height where no cell appears or disappears.

Theorem 4 (the balance law). Let a simple 4-polytope have its cells in two kinds, W and R, with 2 W + 2 R cells at every corner. Then in every generic sweep the swaps that exchange a W–W contact for an R–R contact and those that exchange R–R for W–W are equally many, (R–R faces) − (R cells) + 1, and every other swap exchanges one W–R contact for another. For the bitruncated 600-cell (W: its 600 truncated tetrahedra, the 600-cell's cells cut back; R: its 120 truncated icosahedra, in the places of the 120-cell's cells): 601 each way and 964 mixed.

Proof. With 2 W and 2 R at a corner, the pair that lets go and the pair that meets are complementary: W–W ↔ R–R or W–R ↔ W–R. Every R–R contact appears exactly once in a sweep: at the first corner (one), at the birth of an R cell after the start (one each, R cells − 2 of them) or by a W–W → R–R swap; so these swaps number (R–R faces) − (R cells − 2) − 1; by deaths the same count goes the other way.

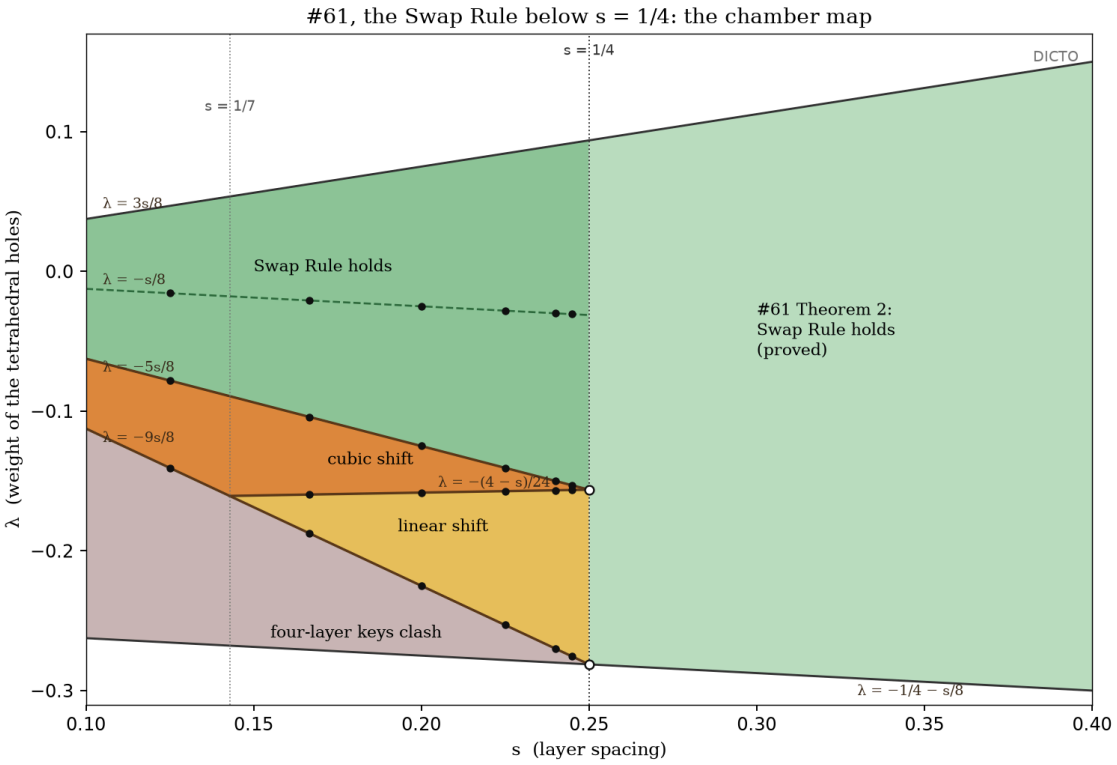
Bitruncated 600-cell: 720 − 118 − 1 = 601; the triangles check it, 1200 = 601 + 598 + 1, and 2166 − 1202 = 964. Confirmed in nine directions. ∎ DICTO's reading: the Queens are the two ends of a channel and this layer, where their cells meet, is where they communicate, equally in both directions.

2. The Swap Rule below s = 1/4

Setup (#61, The Swap Rule). Layers of balls (spacing 1 in a layer) are stacked in a word in A, B, C with layer distance d, $d^2 = (2/3)s$ (s = 1: close packing). Between two layers lie octahedral and tetrahedral holes; space is divided among the holes by power distance $|x - p|^2 - w$, w = 0 for an octahedral hole and w = λ for a tetrahedral one, with λ in #61's range $-1/4 - s/8 \le \lambda \le 3s/8$ (for $s \le 1/4$). A layer is c if its two neighbours differ, h if they are equal. A hole's piece in the gap between layers k and k + 1 is keyed by its role (octahedral, or one of the two tetrahedral kinds) and the types of the four layers k − 1, k, k + 1, k + 2; the types of the two inner layers are (t | t′), the two outer ones the context. V(t | t′) is the piece's volume with the context held fixed. #61 proved V(c|h) + V(h|c) = V(c|c) + V(h|h) when $s \ge 1/4$.

Theorem 3. For every s in [1/8, 1/4], in the six stackings ABC, AC, ACBC, ACABC, ACBCAC, ACBABC (which contain all sixteen four-layer contexts), and in every stacking whose pieces are fixed by the four layers round their gap, the Swap defect D = V(c|h) + V(h|c) − V(c|c) − V(h|h) of each role is the same for all four contexts and equals: for $-9s/8 \le \lambda \le -(4 - s)/24$ (when s > 1/7), $\delta_{tet} = -(1 - 4s)^3/(54s)$, $\delta_{oct} = (1 - 4s)^2(72\lambda + 8 + 13s)/(108s)$; for $\max(-9s/8, -(4 - s)/24) \le \lambda \le -5s/8$, $\delta_{oct} = (8\lambda + 5s)^3/(64s)$, $\delta_{tet} = -(8\lambda + 5s)^2(8\lambda + 2 - 3s)/(64s)$; and D = 0, **the Swap Rule holds**, from λ = −5s/8 to 3s/8 (a further wall at −s/8 leaves D = 0). Below −9s/8 a piece reaches two gaps and the four layers do not fix it (open).

Proof (exact, computer-assisted). s enters only the vertical metric, so every power-diagram plane has a normal linear in s and an offset linear in s and λ; for fixed s only the right-hand sides carry λ, so corners and slacks are affine in λ, and a slack is non-negative on a cell of the map iff it is on the cell's two bounding walls: one-variable rational functions of s. On each cell of the map, built at one sample point for the six words, every corner was re-solved symbolically over Q(s) and every plane's slack proved non-negative on the whole cell (Bernstein sign tests over the s-interval, exact root counting otherwise; about 11 million checks); the volumes and D(s, λ) agree with the closed forms as rational identities. *Neighbourhood.* The cells were built from all sites within squared distance 24 (every plane's slack proved non-negative on the whole cell). No farther site can cut a cell: for any point x of a hole's cell, the octahedral holes of the nearest gap form a triangular lattice of spacing 1 at vertical offset ≤ d/2, so the nearest of them, p_k, has $|x - p_k|^2 \le 1/3 + 1/24 = 3/8$, and hence $|x - p_i|^2 \le 3/8 + |\lambda| < 0.67$: every cell lies within ρ < 0.82 of its site, and $(\sqrt{24} - 0.82)^2 - 0.82^2 > 15.9$ exceeds every weight difference $|\lambda| \le 0.29$. The walls were first found exactly along λ (the Hopscotch walk: each wall is a root where a corner meets a plane) at six spacings. ∎



Dots: walls found exactly by the Hopscotch walk at s = 1/8, 1/6, 1/5, 9/40, 6/25, 49/200; lines through them fitted. All cells close at s = 1/4.

Figure 1. The chamber map of the Swap Rule below s = 1/4. Grey: four-layer keys clash; yellow: linear shift; orange: cubic shift; green: the Swap Rule holds; pale green, s ≥ 1/4: #61's Theorem 2. All cells close at s = 1/4; the walls are proved cell boundaries for every s in [1/8, 1/4] (the dots mark where they were first found).

What happens at the walls. The λ-stack of the 3D power diagrams is a 4D cell complex; in a generic weight walk its 3D slices change by the 3D T1 moves of Theorem 1′ for d = 5, one contact gained or one lost at a time. At #61's walls the lattice's central symmetry forces a loss and a gain onto one point: six cells meet there and the four exchanging holes lie in one plane with a common midpoint (a degenerate 4-4 flip, the rosette kind of event, not Theorem 1's pure swap); in that plane the exchange is a 2D scutoid swap. At s = 1/5, summed over the six words' unit cells, the faces exchanged are: at −(4 − s)/24, 57 between octahedral holes in one layer for 57 between tetrahedral holes in one layer; at −5s/8, 21 between octahedral holes two layers apart for 21 tetrahedral; at −s/8, 126 for 126 between neighbouring layers; at −9s/8, 31 long octahedral contacts lost and 173 gained. The Swap Rule switches on at the moment the last octahedral contact across a whole layer lets go.

3. One dimension up: the 4D cells of the holes

Each hole's power cell, stacked over λ in an interval $[\lambda_a, \lambda_b]$, is a convex 4-polytope $\{(x, \lambda) : n \cdot x \le b_0 + b_1\lambda, \lambda_a \le \lambda \le \lambda_b\}$ whose 3D slices are the cells. We propose the name *4D scutoid* for such a 4-cell whose 3D slices change neighbours at a wall where cells stop touching and others start. At s = 1/5, across the wall λ = −1/8, 25 holes (of the six words) have 4D cells with faces that end or begin exactly at the wall: the octahedral cells' faces towards octahedral holes two layers away end there, the tetrahedral cells' new faces begin there (Figure 2).

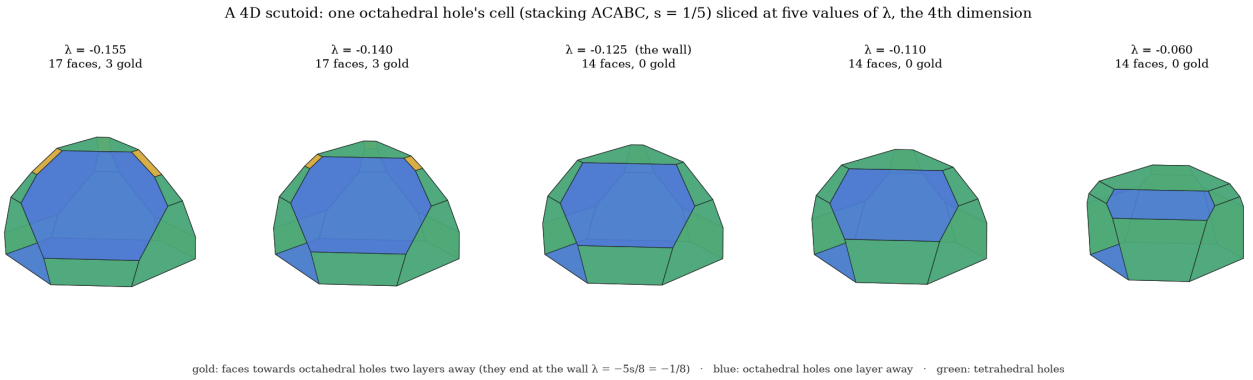


Figure 2. A 4D scutoid: one octahedral hole's cell (stacking ACABC, s = 1/5) sliced at five values of λ. Gold: faces towards octahedral holes two layers away; they end at the wall λ = −1/8 (17 faces before, 14 after).

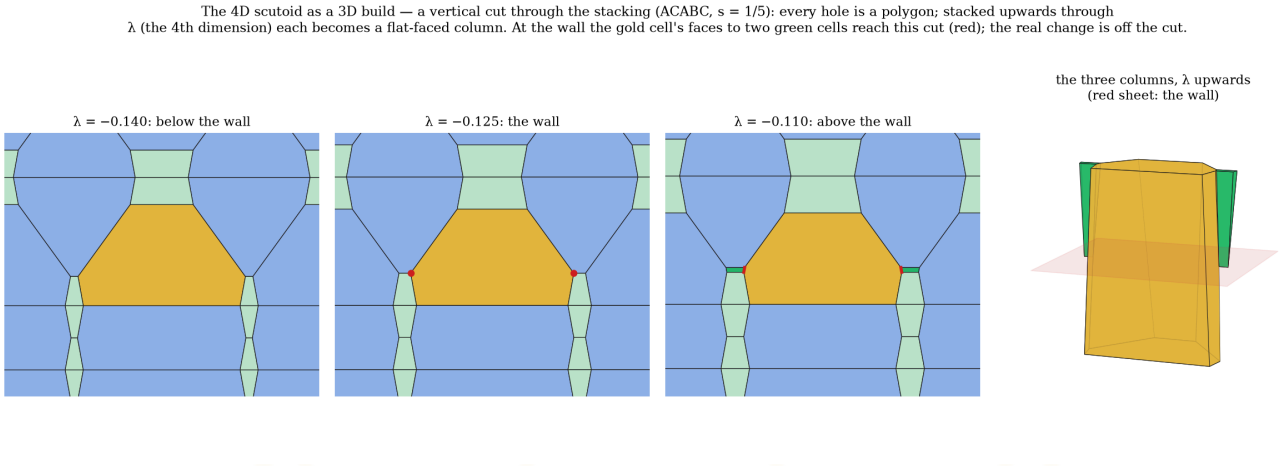


Figure 3. A vertical cut through the stacking (the plane of a set of exchanging holes), every hole a polygon, stacked through λ a flat-faced column. At the wall the two octahedral cells above and below the gold one (two layers apart) stop touching, and tetrahedral holes start touching away from this cut; the gold cell's faces towards the green cells, present throughout, reach the cut here, which is why the green columns appear as wedges at the wall.

Sources checked

Gómez-Gálvez, Vicente-Munuera, Escudero et al., Scutoids are a geometrical solution to three-dimensional packing of epithelia, Nature Communications 9 (2018) 2960; Blankenship, Backovic, Sanny, Weitz and Zallen, Multicellular rosette formation links planar cell polarity to tissue morphogenesis, Developmental Cell 11 (2006) 459–470; Weaire and Rivier, Soap, cells and statistics, Contemporary Physics 25 (1984); Weaire and Hutzler, The Physics of Foams (1999); MathWorld and Wolfram, the scutoid (3D only); McMullen (the h-vector); Ziegler, Lectures on Polytopes, §8.2; Buchstaber and Panov, Toric Topology, ch. 1 (arXiv 1102.1079); Pachner (bistellar moves, 1991); Lickorish, Simplicial moves on complexes and manifolds (Geometry & Topology Monographs 2, 1999); Edelsbrunner and Shah, Incremental topological flipping works for regular triangulations (Algorithmica 1996); Gelfand, Kapranov and Zelevinsky (secondary polytopes); Aurenhammer, Power diagrams (SIAM J. Comput. 1987); Santos, Geometric bistellar flips (arXiv math/0601746); Buchstaber invariant of simple polytopes (arXiv 0908.3407); The Best Ways to Slice a Polytope (arXiv 2304.14239); Izmestiev, Klee and Novik, Simplicial moves on balanced complexes. Web searches (10 Nov 2026): "scutoid four-dimensional 4D polytope slices T1"; "slicing simple 4-polytope hyperplane sweep bistellar flip"; "power diagram weight parameter secondary fan scutoid"; "scutoid polytope higher dimensions"; the rosette paper. No source found reads the sweep of a polytope as a

scutoid tissue or links scutoids to the h-vector. DICTO, The Swap Rule (#61), The Rule of Halves (#57); DICTO studies of 9–10 November 2026.